UCS503 - Software Engineering

GATED: Apartment Tracker

A Smart Digital Platform for Apartment and Society Management

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Chapter 1

Introduction

1.1 Background

Apartment and residential society management involves residents, society administrators, security guards, and maintenance staff. Many societies still depend on manual registers, phone calls, and messaging applications for managing visitors, maintenance requests, and communication.

GATED: Apartment Tracker is a mobile application designed to provide a centralized platform for managing day-to-day activities within an apartment society.

The system connects residents, society administrators, security guards, and maintenance staff through role-based access.

1.2 Problem Statement

Traditional apartment management systems often rely on manual and disconnected processes. Visitor information may be maintained in physical registers, maintenance complaints may be communicated through calls or messages, and residents may have no centralized method to track their requests.

Security personnel also require an efficient way to verify visitors, while society administrators need a centralized system to manage residents and society activities.

Therefore, GATED: Apartment Tracker aims to provide a centralized digital solution for apartment management, visitor access, maintenance tracking, and communication.

1.3 Project Objectives

1. Provide a centralized platform for apartment society management.
2. Implement role-based access for residents, society administrators, security guards, and maintenance staff.
3. Allow residents to register, log in, and access society services.
4. Enable residents to raise and track maintenance requests.
5. Allow security personnel to manage visitor access.
6. Provide administrators with centralized control over society activities.

7. Maintain secure user and society data.

Chapter 2

Methodology and Design

2.1 System Architecture

GATED: Apartment Tracker follows a client-server architecture. The mobile application acts as the client and communicates with the backend for authentication and database operations.

The system uses role-based access control. After authentication and society selection, users are directed to functionality according to their assigned role.

2.1.1 System Components

The major components of the system are:

- **Resident Module:** Provides access to society services, maintenance requests, and visitor management.
- **Society Admin Module:** Provides administrative control over society activities and residents.
- **Security Module:** Handles visitor verification and access management.
- **Maintenance Module:** Handles maintenance requests raised by residents.
- **Authentication Module:** Handles registration and login.
- **Database Module:** Stores users, societies, apartments, maintenance requests, and visitor-related information.

2.1.2 Application Flow

Registration → Login → Society Selection → Role Verification → Role-Based Dashboard

2.2 Use Case Diagram

The Use Case Diagram represents the interaction between the four major actors of the system and the functionalities provided by GATED: Apartment Tracker.

The major actors are Resident, Society Admin, Security, and Maintenance Staff. Each actor has access to functionality according to their role.

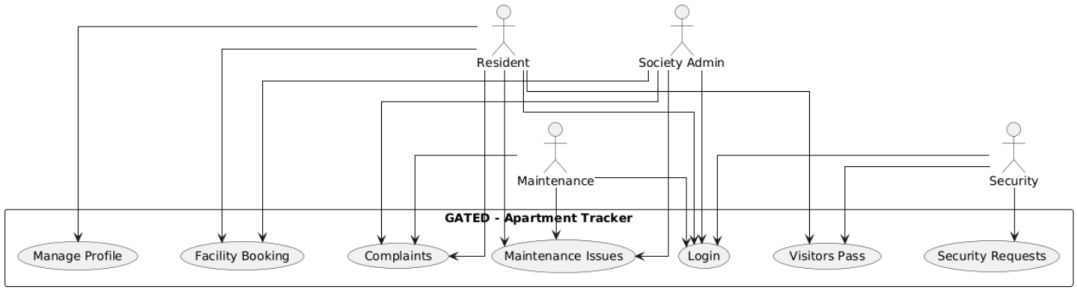


Figure 2.1: Use Case Diagram of GATED: Apartment Tracker

2.3 UML Class Diagram

The UML Class Diagram represents the main software entities and their relationships within the application.

The classes represent users and their roles along with society, apartment, maintenance, visitor, and access management functionality.

2.4 Entity Relationship Diagram

The Entity Relationship (ER) Diagram represents the database structure of GATED: Apartment Tracker. It illustrates the major entities, their attributes, and the relationships between them.

The database primarily stores information related to users, societies, apartments, maintenance requests, visitors, and visitor passes.

2.5 UML Sequence Diagram

The Sequence Diagram represents the interaction between the resident, mobile application, authentication service, database, and other system components during the main application workflow.

The diagram shows the sequence of interactions involved in authentication, society selection, role verification, and access to role-specific functionality. It also illustrates how requests are processed between the mobile application and the backend services.

2.6 Technical Stack

- **Framework:** React Native with Expo
- **Languages:** JavaScript and TypeScript

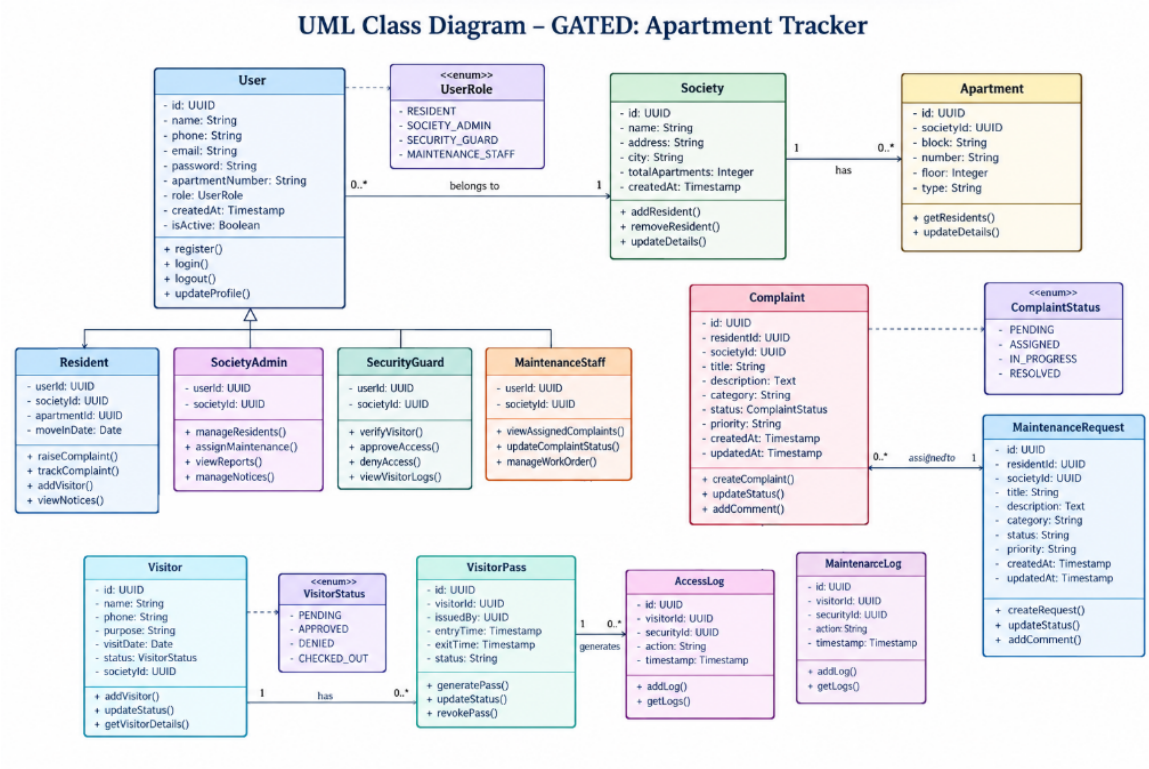


Figure 2.2: UML Class Diagram

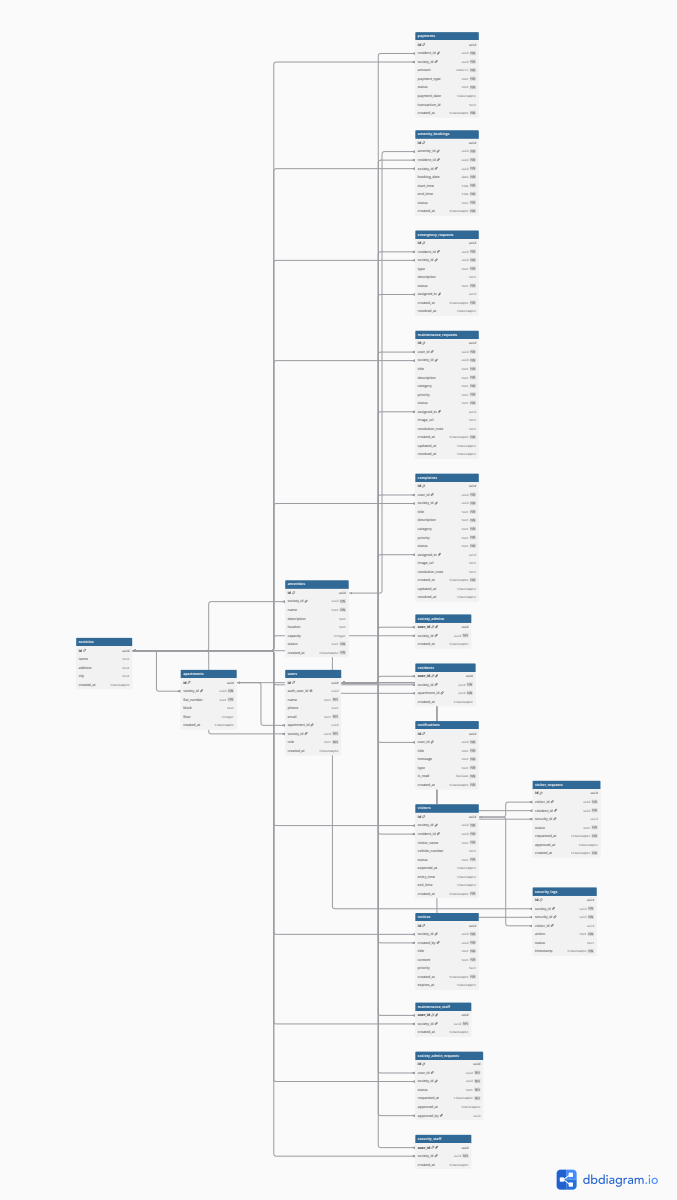


Figure 2.3: Entity Relationship Diagram

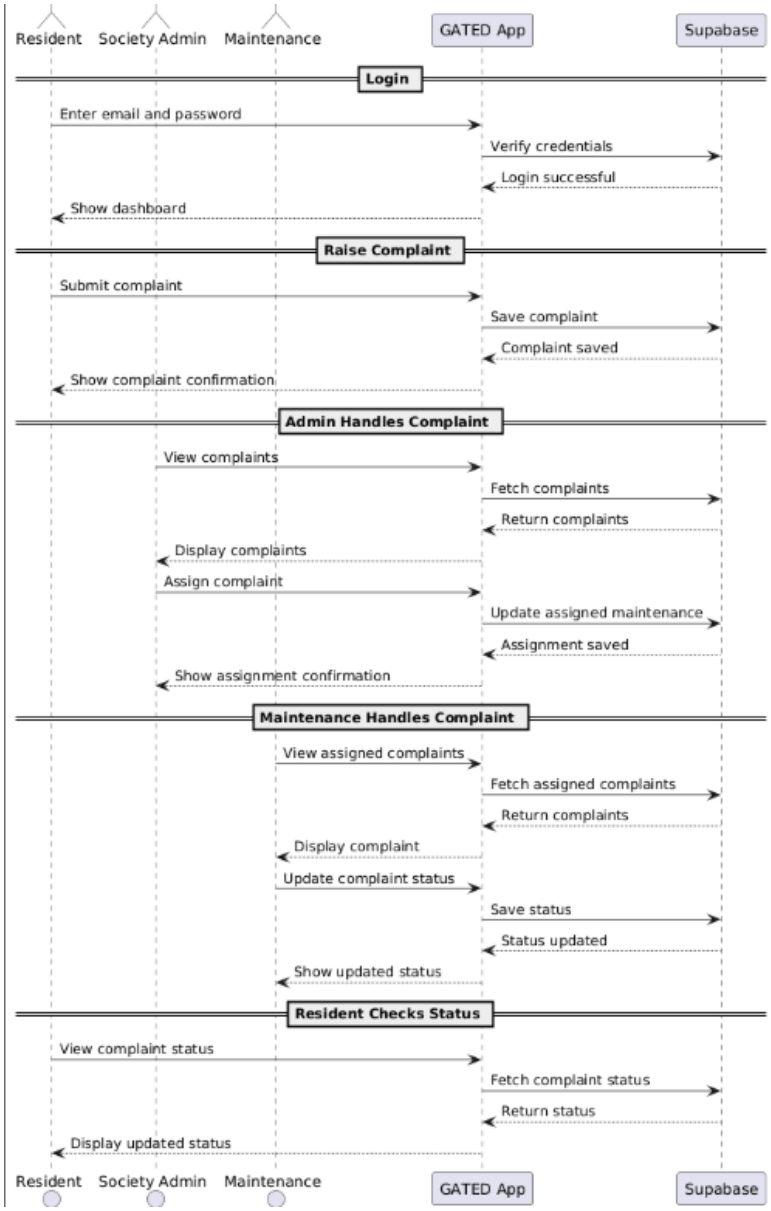


Figure 2.4: UML Sequence Diagram

- **Database:** PostgreSQL through Supabase
- **Backend:** Supabase
- **Authentication:** Supabase Authentication
- **Navigation:** Expo Router
- **Development Tools:** Visual Studio Code, Expo Go, Git and GitHub
- **Security:** Supabase Row Level Security

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2.7 Implementation

2.7.1 Authentication and Role Management

The authentication module provides secure registration and login functionality using Supabase Authentication. During registration, residents provide their name, phone number, email address, apartment number, and password. The authentication system verifies the user's credentials and allows only authenticated users to access the application.

After successful login, the user selects their registered society. The system then retrieves the user's role and directs them to the appropriate dashboard. Role-based access ensures that each type of user can access only the features relevant to their responsibilities.

The supported roles are:

- **Resident:** Can manage their profile, raise complaints, track maintenance requests, and manage visitor-related activities.
- **Society Admin:** Can oversee residents, complaints, maintenance activities, and other society-level operations.
- **Security Guard:** Can verify visitors and manage visitor entry and access within the society.
- **Maintenance Staff:** Can view assigned complaints, work on maintenance issues, and update their status.

2.7.2 Complaint and Maintenance Management

The Complaint and Maintenance Management module provides a structured system for reporting and resolving maintenance-related issues. Residents can raise a complaint by providing details such as the complaint title, description, category, and other relevant information.

Once submitted, each complaint is stored in the database and associated with the resident who created it. This allows the complaint to be tracked throughout its lifecycle. The system maintains different statuses such as *Pending*, *Assigned*, *In Progress*, and *Resolved*.

Society administrators can view submitted complaints and assign them to the appropriate maintenance staff. Maintenance personnel can then view the complaints assigned to them, work on the reported issue, and update its status as the work progresses.

Residents can track the current status of their complaints, providing transparency about whether an issue has been received, assigned, is being worked on, or has been resolved. This reduces dependency on phone calls and messaging applications for following up on complaints.

The system also maintains a record of previous complaints, which can help administrators monitor recurring issues, identify common maintenance problems, and improve the overall maintenance process of the society.

2.7.3 Visitor Management

The Visitor Management module helps manage visitor access to the apartment society. Residents can provide visitor information when required, while security personnel can verify the visitor's details at the society entrance.

Security guards can review the available visitor information and approve or deny entry according to the access details provided. This provides a more organized alternative to maintaining visitor information through manual registers.

The module also maintains visitor-related records, making it easier for security personnel and administrators to monitor visitor activity within the society.

2.7.4 Society Administration

The Society Administration module provides society administrators with higher-level access to manage society-related activities. Administrators can oversee residents, monitor complaints, assign maintenance requests, and supervise the overall functioning of the platform.

The administrator dashboard provides a centralized view of important society operations and helps coordinate activities between residents, security personnel, and maintenance staff.

2.7.5 Data Security

Data security is implemented using Supabase Authentication and Row Level Security (RLS). Authentication ensures that only registered users can access the application, while RLS policies restrict database operations according to the user's role and permissions.

For example, residents should only be able to access information relevant to their own account and activities, while maintenance staff can access complaints assigned to them. Administrative operations are restricted to users with the appropriate administrative permissions.

This role-based security approach helps prevent unauthorized access and protects sensitive resident, society, complaint, and visitor information.

Chapter 3

Results and Evaluation

3.1 Testing

The application is tested through functional and integration testing.

The major workflows tested include:

- User registration
- User login
- Society selection
- Role-based navigation
- Complaint request creation
- Complaint request management
- Database operations
- Role-based access restrictions

3.2 Analysis

The implementation demonstrates that a mobile-based platform can centralize multiple apartment society management activities.

The role-based architecture separates the responsibilities of residents, society administrators, security guards, and maintenance staff.

Supabase provides authentication and database services, while Row Level Security provides additional protection against unauthorized database access.

During development, challenges related to authentication, database constraints, role assignment, navigation, and Row Level Security were addressed through iterative testing and configuration.

Chapter 4

Conclusion

4.1 Summary

GATED: Apartment Tracker is a mobile application designed to simplify and centralize apartment society management.

The system connects residents, society administrators, security guards, and maintenance staff through a role-based platform.

The application provides authentication, society selection, role-based dashboards, maintenance request management, and visitor access management.

React Native with Expo is used for mobile application development, while Supabase provides authentication and database services.

4.2 Future Scope

The system can be further enhanced with:

1. Real-time notifications for visitors and maintenance requests.
2. QR-based visitor passes.
3. Advanced delivery and visitor management.
4. Emergency alerts and emergency contacts.
5. Society-wide announcements.
6. Online maintenance fee and payment management.
7. Analytics dashboards for administrators.
8. Support for multiple societies and larger numbers of users.

4.3 Final Remarks

GATED: Apartment Tracker demonstrates how mobile application development and cloud-based backend services can be used to improve residential society management.

The proposed system provides a centralized platform for residents, administrators, security personnel, and maintenance staff. With further development, the application can be extended into a comprehensive smart apartment management solution.